EEP 595: Privacy-Preserving Machine Learning – Spring 2026
Course Project - Overview

Assigned: Wednesday, April 8, 2026
Instructor: Tamara Bonaci
Department of Electrical and Computer Engineering
University of Washington

1 Overview

Over the course of the next nine weeks, you will explore a topic of your choice at **the intersection of privacy and machine learning**. Your project **needs to** combine two equally important components:

- **A literature review component** You are expected to identify, read, and critically engage with relevant research papers. This is not a superficial survey - you should understand the key ideas, situate them in the broader PPML landscape, and be able to discuss their strengths, limitations, and open questions.
- **A practical component** You are expected to complement your literature work with something hands-on - **an implementation, a set of experiments, a simulation, or a prototype system.** Purely theoretical or survey-only projects will not meet the requirements of this course.

Both components should be clearly present in every phase of the project, from the proposal through the final report. *A strong project is one where the practical work is informed by the literature, and the literature is illuminated by the practical work.*

Your projects can be individual or in teams of up to three people. Team size will be factored into expectations - larger teams are expected to produce broader, more comprehensive work.

Possible directions include, but are not limited to:

- Implementing and evaluating a PPML technique (e.g., federated learning, differential privacy, secure multi-party computation, homomorphic encryption) on a real-world dataset.
- Comparing the privacy-utility tradeoff of multiple approaches on a shared problem.
- Identifying a privacy vulnerability in an existing ML pipeline, and proposing/designing a mitigation.
- Reproducing and critically extending the results of a published PPML paper.
- Designing and prototyping a system that applies PPML techniques to a specific application domain (e.g., healthcare, finance, NLP).

2 Part 1 - Team Formation, Project Proposal and Planning

In the next two wees, please form your team, or decide to work alone, and settle on a topic. This project part has two components, which your will want to submit together as a single document.

2.1 Component 1: Project Proposal (2 pages)

Think of your proposal as a pitch to a results-oriented stakeholder - a potential client or executive - who may or may not be deeply familiar with the problem. Get to the point quickly, and make the case that the problem matters and that your approach is sound.

Your proposal should address the following:

- **1. Problem description** Please provide a clear description of the PPML problem you are addressing. If your project focuses on a specific attack or class of attacks, you will likely want to structure your analysis around:
  - **System:** What is it, and how does it work?
  - **Assets:** What are you trying to protect, and why? How valuable are those assets?
  - **Attackers:** Who might attack the system, and what would motivate them?
  - **Vulnerabilities:** Where might the system be weak?
  - **Threats:** What actions might an attacker take to exploit those weaknesses?
  - **Risks:** How likely is exploitation, and how severe would the consequences be?

Diagrams and figures are welcome but not required.

- —bf 2. Proposed approach If your project focuses on preventing or mitigating the described problem, describe your planned approach. Consider addressing:
  - What broad category does your solution fall into (e.g., cryptography, system redesign, differential privacy)?
  - What are the expected benefits of the solution?
  - What are the known or anticipated drawbacks?
- **3. References.** For each paper you are referencing or plan to reference, provide the full title and author list.

2.2 Component 2: Project Plan (1 page)

Alongside your proposal, please submit a one-page project plan. This is not a SCRUM document — it is simply a structured, honest account of what you intend to do and when. It should include:

- **Final product description.** A brief list of the expected deliverables at the end of the project (report, presentation, code, experiments, etc.) and what each will contain.
- **Milestone plan.** A week-by-week breakdown of what you plan to accomplish during this project. Be concrete - vague entries like ”work on implementation” are less useful than “complete baseline federated learning prototype and run initial accuracy benchmarks.”
- **Responsibilities.** If working in a team, indicate which team member is leading each major workstream.

We will review proposals and return comments promptly so you can begin your work with confidence.

2.3 Grading Criteria

We will grade your proposals based upon:

- Clarity and significance of the research question.
- Quality of threat/problem analysis.
- Demonstrated awareness of relevant prior work.
- Feasibility and concreteness of the plan.
- Appropriateness of scope given team size.

3 Part 2 - Initial Progress (Weeks 4 - 7)

By the end of this phase, you should have made concrete progress on both the technical and conceptual sides of your project - for example, a working baseline implementation, preliminary experimental results, or a detailed analysis of the techniques you are studying.

3.1 Deliverable

As a result of this phase of the project , you will want to submit a **2-page progress report**. You will want to report on the following:

Progress and execution:

- What have you accomplished so far in your analysis, threat modeling (if applicable), or development of mitigation and prevention strategies (if applicable)?
- What obstacles or challenges have you encountered, particularly in the analysis, threat modeling, or any coding and simulation work?
- What harder steps or challenges do you still anticipate, and how do you plan to address them?

Engagement with the literature: For the most relevant papers you are studying, researching, or extending, consider reflecting on:

- What important ideas and concepts have you learned from the paper, why are they significant, and how do they relate to the course?
- Are there challenging or unfamiliar concepts in the paper that have required additional reading? If so, what resources have you used?

- Do you have any technical concerns about the paper - for example, missing steps in derivations, or an incomplete experimental setup - and how do you plan to investigate them?
- How do you plan to present these ideas to the class?

3.2 Grading Criteria

We will grade your progress reports based upon:

- Tangible, demonstrable progress relative to the proposal and plan.
- Quality of any preliminary results or implementation work.
- Honest and thoughtful reflection on challenges encountered.
- Depth of engagement with the literature.
- Credibility of the remaining plan.

4 Draft Project Report (Weeks 7–9)

During this phase of your project, your will want to complete technical work of your project, and write a draft of your project report. This draft will be reviewed by the instructor and by several of your peers. You are want to take their feedback seriously and incorporate it into your final submission.

4.1 Deliverable

A **draft** final report submitted at the end of Week 9. Emphasis on the word **draft** - this does not need to be a polished, complete document. It is expected to have rough edges, open questions, and sections still in progress. What it must demonstrate is that your project is substantively underway: your core argument should be clear, your practical work should be described and at least partially complete, and your engagement with the literature should be evident.
The draft will be reviewed by the instructor and by several of your peers. Peer reviewers will be asked to comment on clarity, technical soundness, and completeness. You will receive their written feedback before the final submission deadline, and you are expected to take it seriously and incorporate it into your final report.
There is no strict page limit, but a typical solid draft for a team of three might run 6–8 pages. Structure it however best suits your project, but it should cover at minimum:

- **Introduction** - motivation, problem statement, and contributions.
- **Background and related work** - the PPML landscape relevant to your topic.
- **Approach** - technical description of what you implemented or analyzed.
- **Evaluation** - experimental results, case studies, or analytical findings, even if preliminary, with honest discussion of limitations.
- **Conclusion** — what you have learned so far, and what remains.

4.2 Grading Criteria

The grade for this phase has two equal components:

- **Your draft** - evaluated on completeness relative to its draft status, technical depth, quality of literature engagement, and clarity of writing.
- **Your peer reviews** - evaluated on thoroughness, constructiveness, and the quality of technical feedback you provide to others. Simply saying a draft is ”good” or ”needs more work” is not sufficient — reviewers are expected to engage with the content in detail.

5 Final Presentation and Project Report (Final Week)

At this point, you are ready to deliver a presentation of your project and submit your revised final report incorporating all feedback received.

5.1 Presentation

Each team will present for **8–10 minutes**), followed by a few questions. The presentation should be accessible to the rest of the class - assume your audience has taken the same course, but is not an expert on your specific topic. A demo is strongly encouraged where applicable.

5.2 Final Project Report

Please submit a revised version of your draft report. A brief appendix, noting what changes were made in response to peer and instructor feedback is required.

5.3 Grading Criteria

- **Presentation:** clarity, depth, quality of any demo, and handling of questions.
- **Final report:** improvement over the draft, incorporation of feedback, and overall technical quality and writing.

6 A Note on Scope and Ambition

Graduate-level work is expected. That does not mean your project needs to be groundbreaking, but it does mean engaging seriously with the technical and ethical dimensions of your topic. Privacy-preserving ML involves real tradeoffs — between utility and privacy, between practicality and theoretical guarantees, between individual and collective interests. The strongest projects will grapple with these tradeoffs honestly rather than treating them as footnotes.